Internship & Project Collaboration Platform

UI/UX DESIGN, APPLIED PRINCIPLES & ARCHITECTURE REPORT

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Course: Software Engineering • DA-2
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1. Project Overview & UI/UX System Architecture

The **Internship and Project Collaboration** platform is a centralized, role-based web application engineered to bridge students, faculty mentors, and hiring organizations. It streamlines opportunity discovery, multi-disciplinary capstone collaboration, milestone evaluation, and talent portfolio management.

> **Report Objectives & Scope:** This document provides the formal software engineering and UI/UX report for Digital Assignment 2 (DA-2). It encompasses all 6 high-fidelity Figma UI screens, a detailed breakdown of **Applied Design Principles** (*Abstraction, Modularity, High Cohesion, Low Coupling*), and an analytical justification of **5 Key Design Decisions**.

1.1 User Personas & System Access Tiers

Role / Persona	Primary Capabilities & Workflows	Assigned UI Screens & Views
Student	Explore verified internships, apply via 1-click workflows, manage capstone Kanban sprint tasks, track ATS resume score, and collaborate with peers.	Screen 01 (Auth), Screen 02 (Dashboard), Screen 03 (Internships), Screen 04 (Projects), Screen 05 (Profile).
Faculty / Mentor	Supervise student capstones, review uploaded architecture documents, grade milestones (Approve / Request Revision), and monitor class velocity.	Screen 01 (Auth), Screen 06 (Faculty Supervision Dashboard), Review & Grading Modal.
Recruiter / Org	Post verified opportunities, filter applicant pools by verified skills and GPA, review candidate portfolios, and manage candidate pipelines.	Screen 01 (Auth), Screen 03 (Talent & Opportunity Matcher).

1.2 Client UI Architecture & Layering

The front-end is architected as a lightweight, high-performance **Single-Page Application (SPA)**. The system separates concerns into three decoupled tiers:

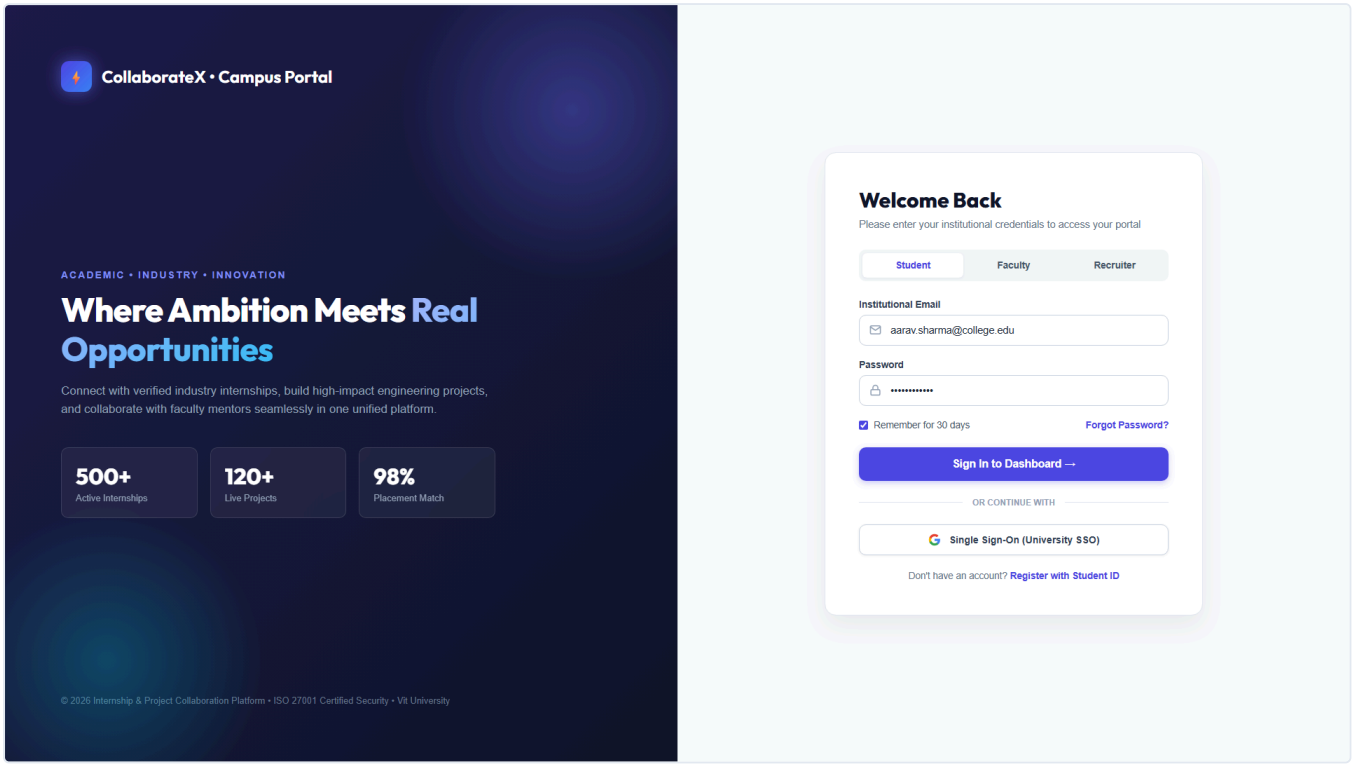
- **Presentation Layer:** Modular HTML5 semantic views styled with semantic CSS design tokens.
- **Abstraction & Event Bus Layer:** Decoupled pub/sub event dispatcher (`EventBus`), centralized state repository (`AppStore`), declarative client-side route manager (`NavigationRouter`), and toast notification service (`NotificationService`).
- **Data & Entity Layer:** High-cohesion domain repositories for Users, Internships, Capstone Projects, and Faculty Evaluation Queues.

2. High-Fidelity UI Screens (Screens 01 & 02)

AUTHENTICATION, ROLE SELECTION & STUDENT DASHBOARD

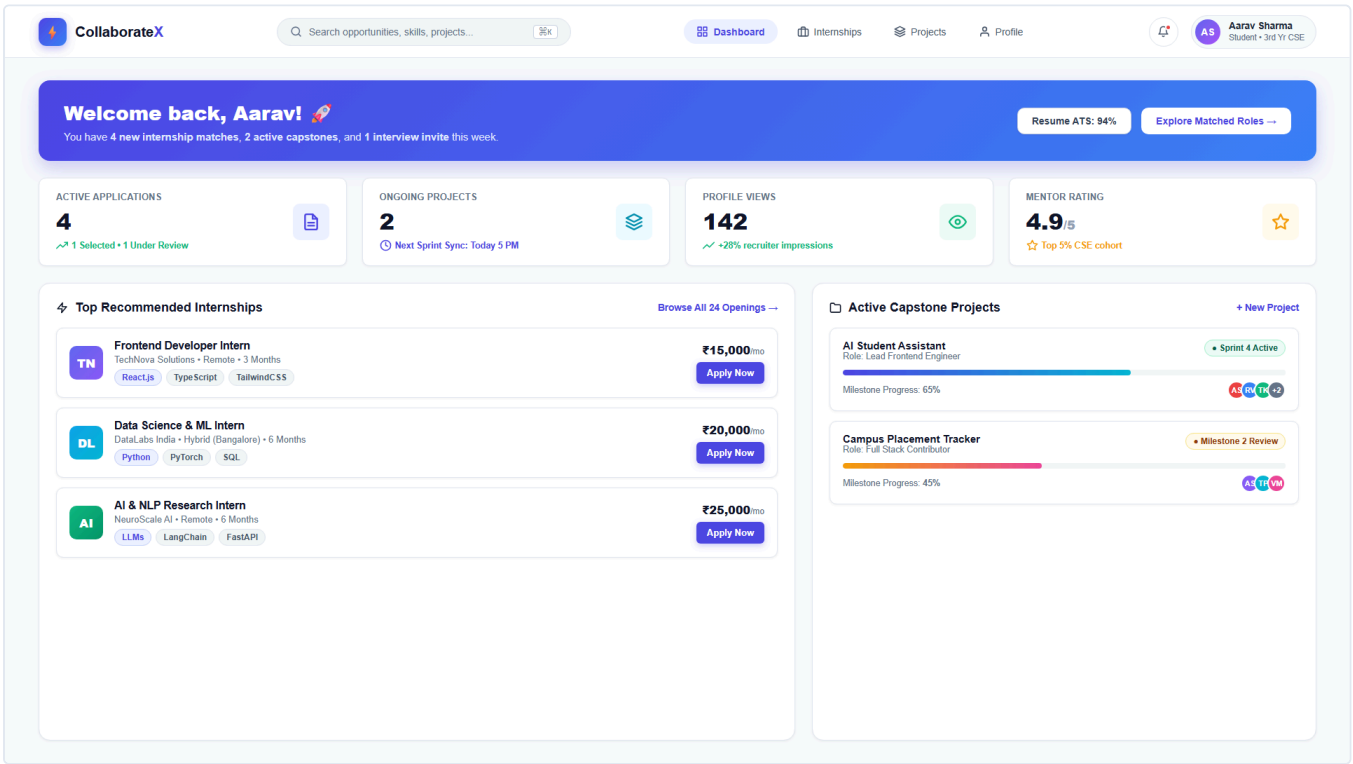
Student: Srishti Jain (24BRS1412)
Format: High-Resolution 1920×1080 Mockups

Screen 01 • Secure Authentication & Institutional Role Selector



Key Features & UX Highlights: Modern SaaS split-screen layout with dark mesh branding; dynamic 3-role toggle (Student / Faculty / Recruiter); institutional University Single Sign-On (SSO) button; floating inputs; ISO 27001 trust indicator.

Screen 02 • Student Dashboard & Opportunity Recommendation Hub



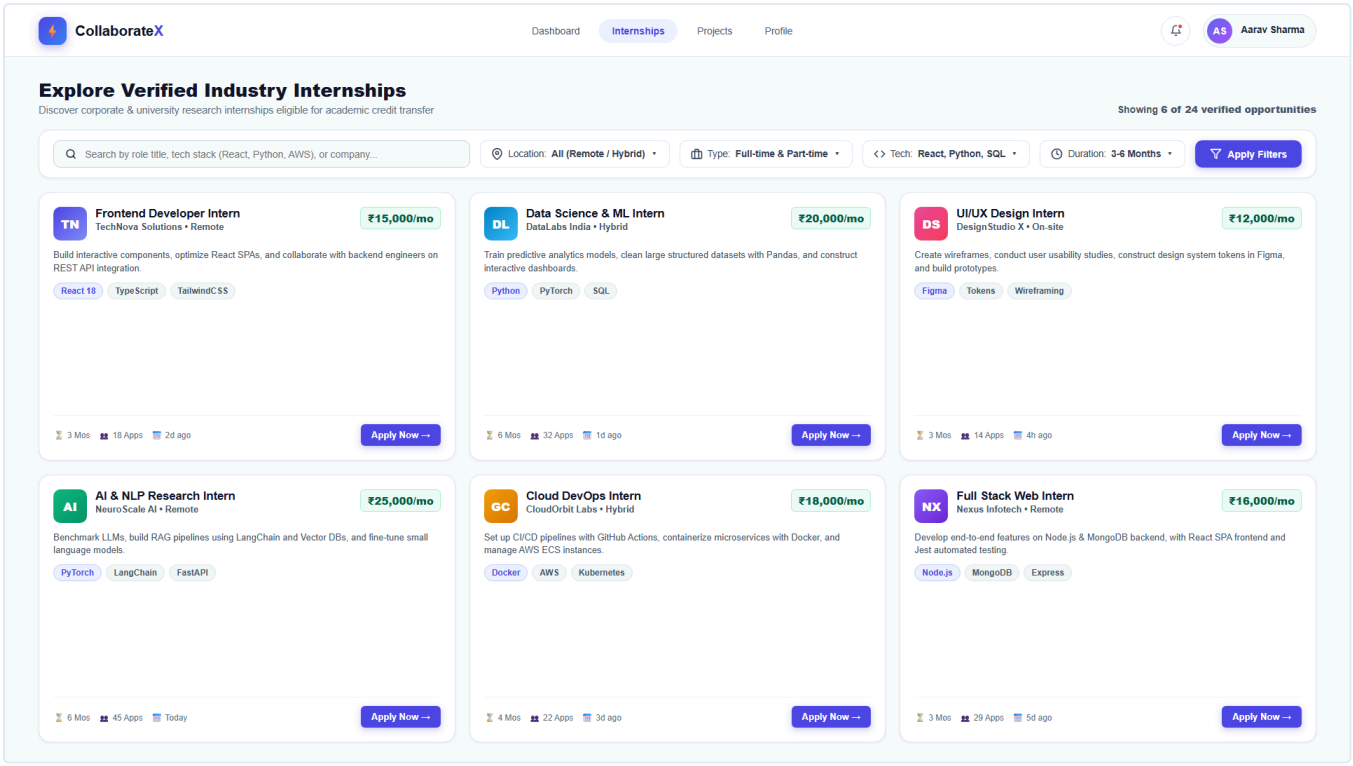
Key Features & UX Highlights: Global header with ⌘K search palette; 4 primary KPI metrics (Active Applications, Projects, Profile Views +28%, Mentor Score 4.9/5); personalized greeting; recommended internship cards with stipend tags; active capstone progress meters.

2. High-Fidelity UI Screens (Screens 03 & 04)

BROWSE INTERNSHIPS & PROJECT COLLABORATION WORKSPACE

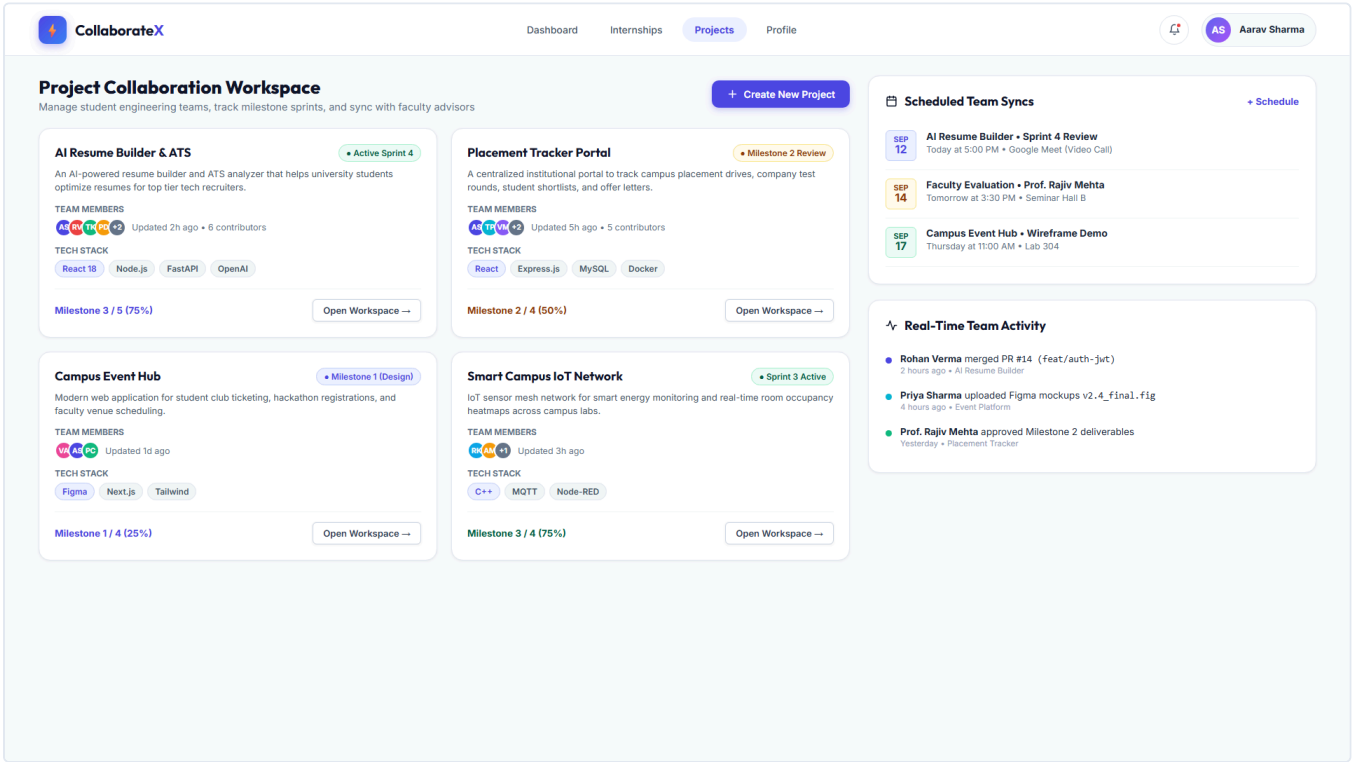
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Format: High-Resolution 1920×1080 Mockups

Screen 03 • Browse Internships & Faceted Multi-Tag Filtering



Key Features & UX Highlights: Sticky multi-tag search bar (Location, Work Type, Tech Stack, Duration, Stipend range); 6 rich opportunity cards featuring company logos, verified badges, stipend badges (₹12k - ₹25k/mo), applicant counters, and 1-click modal application drawer.

Screen 04 • Project Collaboration Workspace & Sprint Kanban



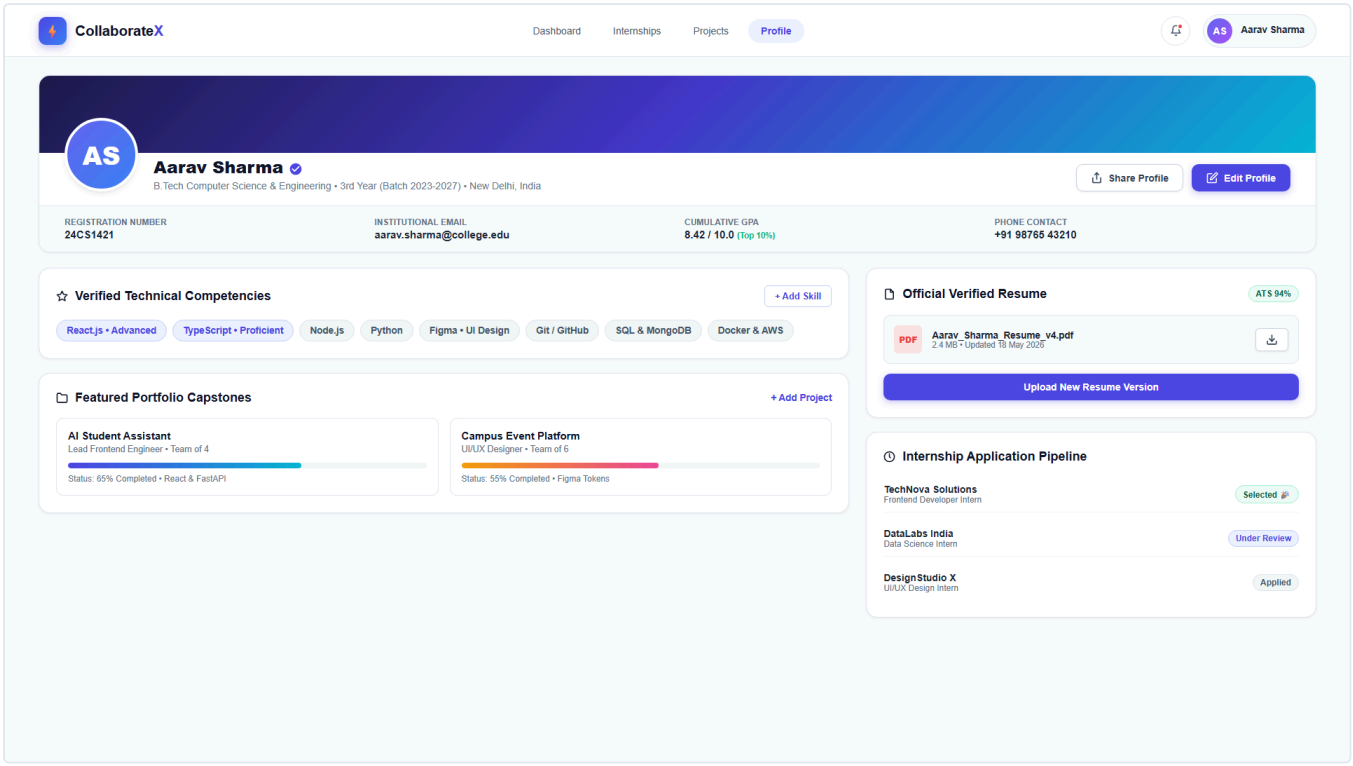
Key Features & UX Highlights: 4 Kanban milestone sprint cards (AI Resume Builder, Placement Tracker, Event Hub, Smart Campus IoT); interactive milestone checklists (8/10 Done); scheduled Google Meet syncs; assigned mentor badges; real-time team activity feed.

2. High-Fidelity UI Screens (Screens 05 & 06)

STUDENT PROFILE PORTFOLIO & FACULTY SUPERVISION DASHBOARD

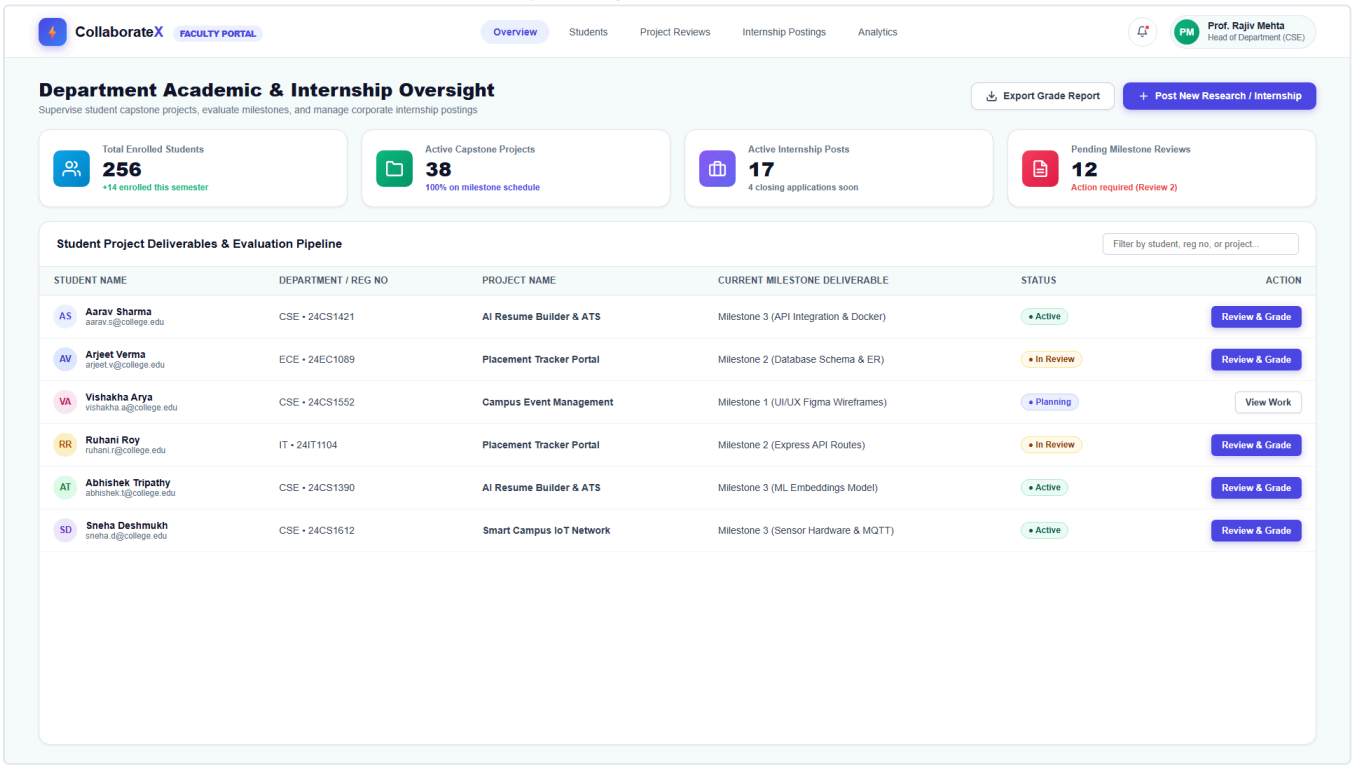
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Format: High-Resolution 1920×1080 Mockups

Screen 05 • Student Profile, Skill Matrix & ATS Resume Portfolio



Key Features & UX Highlights: Gradient cover banner with verified avatar; academic information grid (B.Tech CSE, CGPA 8.42/10.0); categorized technical skills matrix; ATS 94% verified resume manager with upload/download; 3-stage visual application pipeline tracker.

Screen 06 • Faculty Oversight & Milestone Evaluation Dashboard



Key Features & UX Highlights: Institutional analytics cards (256 students, 38 projects, 17 postings, 12 pending reviews); structured student project evaluation table with status pills (Under Review, Approved, Needs Revision); 1-click modal grading and feedback

3. Design Principles Applied (Section a)

ABSTRACTION, MODULARITY, COHESION, AND LOW COUPLING

3.1 Abstraction (Hiding Complexity Behind Clean Interfaces)

Abstraction ensures that higher-level components interact with clear, declarative interfaces while low-level implementation details (data persistence, DOM rendering, network operations) remain encapsulated.

- **Centralized State Repository (`AppStore` in `app/app.js`):** UI components do not perform direct mutations on underlying datasets. Instead, they invoke high-level abstract methods such as `store.applyInternship(id)`, `store.toggleTask(projId, taskIdx)`, or `store.setRole('faculty')`. The inner state transformations and event triggers are completely hidden.
- **Decoupled Pub/Sub Event Bus (`EventBus`):** Modules publish domain events without knowing which subscribers exist, abstracting the communication channel.
- **Toast Feedback Service (`NotificationService`):** Any module requests notifications via `appEventBus.publish('notify', { message, type })`, abstracting away DOM insertion, timer lifecycles, and animation keyframes.
- **CSS Design Tokens (`app/styles.css`):** Visual attributes are abstracted into semantic custom properties (`--primary: #4f46e5`, `--shadow-md`, `--radius-lg`), shielding developers from hardcoded hex codes.

3.2 Modularity (Decomposition into Self-Contained Units)

Modularity divides the software system into distinct, interchangeable modules, each responsible for a specific functional domain.

- **6 Discrete Screen Feature Modules:** The codebase is organized into 6 class-based modules corresponding directly to the Figma designs: `AuthModule`, `DashboardModule`, `InternshipsModule`, `ProjectsModule`, `ProfileModule`, and `FacultyModule`.
- **Atomic UI Components:** Common elements (`.metric-card`, `.status-badge`, `.tag-stipend`, `.kanban-col-card`, `.modal-overlay`) are defined once in CSS and reused across all views.
- **Separation of Architectural Concerns:** Strict segregation between markup (`index.html`), style design system (`styles.css`), and runtime state orchestration (`app.js`).

3.3 High Cohesion (Single, Focused Responsibility)

High Cohesion ensures that all functions and data within a module are tightly related and serve a single, well-defined purpose (Single Responsibility Principle).

- **Domain-Specific Isolation:** `InternshipsModule` only handles filtering, searching, and opportunity rendering. It contains no milestone or grading logic. Conversely, `FacultyModule` is strictly dedicated to managing the review queue and evaluation modals.
- **Pure Data Transformations:** The filter engine inside `InternshipsModule.render()` performs pure filtering operations on in-memory state without side effects on external modules.

3.4 Low Coupling (Minimizing Inter-Module Dependencies)

Low Coupling minimizes inter-dependencies between components so modifications in one unit do not cause unintended ripple effects.

- **No Direct Cross-Module Invocations:** When a user applies for an internship, `InternshipsModule` does not query or mutate DOM elements belonging to `DashboardModule` or `ProfileModule`.
- **Event-Driven Pub/Sub Synchronization:** Applying for an internship emits `internship:applied`. The dashboard counter and toast service listen independently and react without coupling to the caller.
- **Decoupled Business Logic:** `AppStore` contains zero DOM or browser window dependencies, allowing it to execute and be tested in headless environments.

4. Key Architectural & UX Choices (Quick List: 3–5 Key Choices)

Below is the structured analysis of the 5 primary design decisions made during the development of the Internship and Project Collaboration platform, outlining alternatives considered and the technical/UX justifications.

#	Design Decision / Choice	Alternative Considered	Engineering & UX Justification (Why?)
1	**Role-Based Single-Page App (SPA) Architecture**	Multi-Page Website (separate HTML page per screen)	**Zero-Latency Screen Switching:** Eliminates page reload flicker, retains active in-memory search/filter states and Kanban checklist progress across views, and provides desktop-app smoothness for students and mentors.
2	**Centralized Event-Driven Pub/Sub State Model (`EventBus`)**	Direct cross-component DOM updates & global procedural scripts	**Enforces Strict Low Coupling:** When a milestone is completed or an internship applied, KPI cards, activity feeds, and toast notifications synchronize automatically without spaghetti dependencies.
3	**Faceted Multi-Tag Client-Side Filter Engine**	Server-side form submissions requiring page reloads	**Instant Discovery:** Students can filter by Work Type (*Remote / Hybrid / On-site*) and Tech Stack (*React, Python, Docker, Figma*) simultaneously with immediate reactive re-rendering, reducing cognitive effort.
4	**Semantic CSS Design Tokens (Custom Properties)**	Hardcoded inline styles / ad-hoc utility classes	**Design Consistency & Accessibility:** Guarantees visual harmony across all 6 Figma screens, meets WCAG AA contrast standards, and allows seamless future theming (e.g., institution-specific branding or dark mode).
5	**Interactive Faculty Evaluation Modal Workflow**	Static non-interactive PDF feedback sheets / manual emails	**Closes the Feedback Loop:** Enables faculty mentors to inspect milestone submissions and instantly record decisions (*Approved / Needs Revision*) with rubric remarks in real time.

5. Summary of Deliverables & Codebase Assets

Deliverable	Repository Location	Format / Technology
Working UI/UX Application	`app/index.html`, `app/styles.css`, `app/app.js`	Vanilla HTML5, CSS3 Tokens, ES6 JS
High-Res Figma Mockup Screens	`docs/design/screens/*.png` & `docs/figma/`	1920×1080 High-Res PNG
Compiled Academic PDF Report	`UI_UX_Design_Report(TimesNewRoman).pdf`	PDF Document (Times New Roman)